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MAXIMIZING AND MINIMIZING



Goldilocks preferred the middle, but in the world of algorithms we're usually more interested in the extreme highs and lows. Some powerful algorithms enable us to reach maxima (for example, maximum revenue, maximum profits, maximum efficiency, maximum productivity) and minima (for example, minimum cost, minimum error, minimum discomfort, and minimum loss). This chapter covers gradient ascent and gradient descent, two simple but effective methods to efficiently find maxima and minima of functions. We also discuss some of the issues that come with maximization and minimization problems, and how to deal with them. Finally, we discuss how to know whether a particular algorithm is appropriate to use in a given situation. We'll start with a hypothetical scenario—trying to set optimal tax rates